

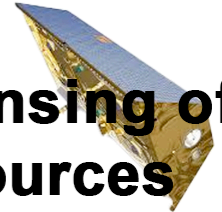


Australian
National
University

Research School
of Earth Sciences



**Remote Sensing of
Water Resources**
EMSC3025/6025



Runoff

EMSC3025/6025

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EMSC3025/6025: Remote Sensing of Water
Resources

Dr. Sia Ghelichkhan

Objectives

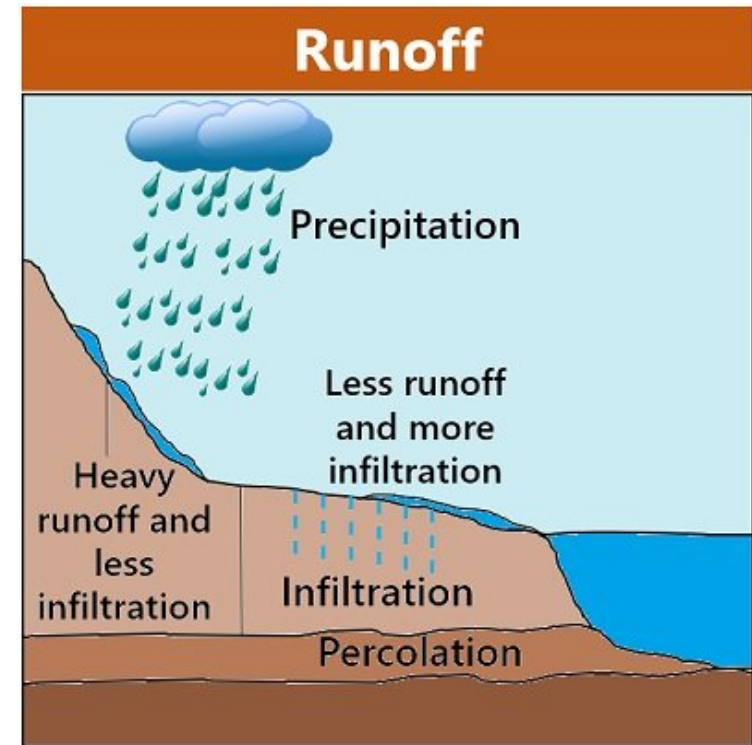
By the end of this lecture, you should be able to:

- Understand the processes that generate runoff and lead to channelized flow.
- Identify and explain the main mechanisms responsible for delivering water to streams and rivers.
- Distinguish between overland flow, subsurface flow, and groundwater contributions to streamflow.
- Describe and compare key theories of stormflow generation (Horton, Betson, Hewlett & Hibbert).
- Explain the role of macropores and piston flow in rapid water movement.
- Apply and evaluate methods for measuring streamflow, including the velocity–area method, stage–discharge relationships, and the use of flumes and weirs.
- Recognise the limitations and considerations of different streamflow measurement techniques.

Runoff

Definitions

- Any process that results in water moving toward channelized flow, following **precipitation reaching the surface**.
- Mechanisms responsible for delivering water to streams and rivers.
- River/streamflow**: The movement of water in channels toward the ocean.
- Riverflow is typically quantified as **discharge**: *volume of water per unit time $\frac{m^3}{s}$* .



biologyreader.com: Components involved in runoff.

Hydrographs and runoff

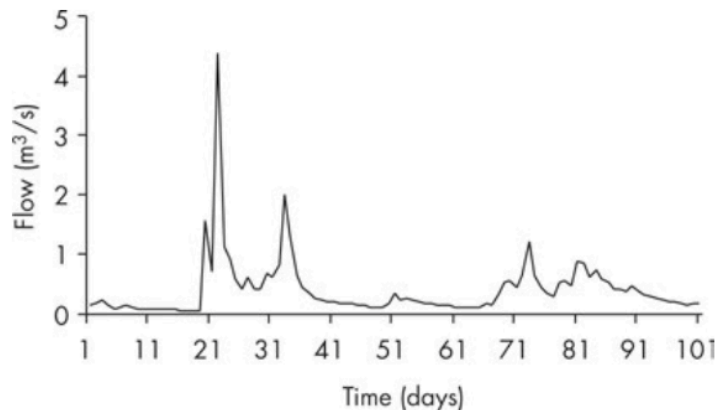


Figure 1: River Wye: Average values for 100 days during autumn of 1995

- A continuous record of streamflow is called a **hydrograph**.
- **Peak/storm flow**: Water present in a stream during and immediately after a significant rainfall event.
- Globally controlled by rainfall intensity and duration; at catchment scale, affected by catchment **size, slope, shape**, soil characteristics, vegetation type and cover, degree of urbanisation, and **antecedent soil moisture**.
- Periods between peaks are referred to as **baseflow** (distinct from low flow), typically understood to be supplied by groundwater.

Catchments and their geology

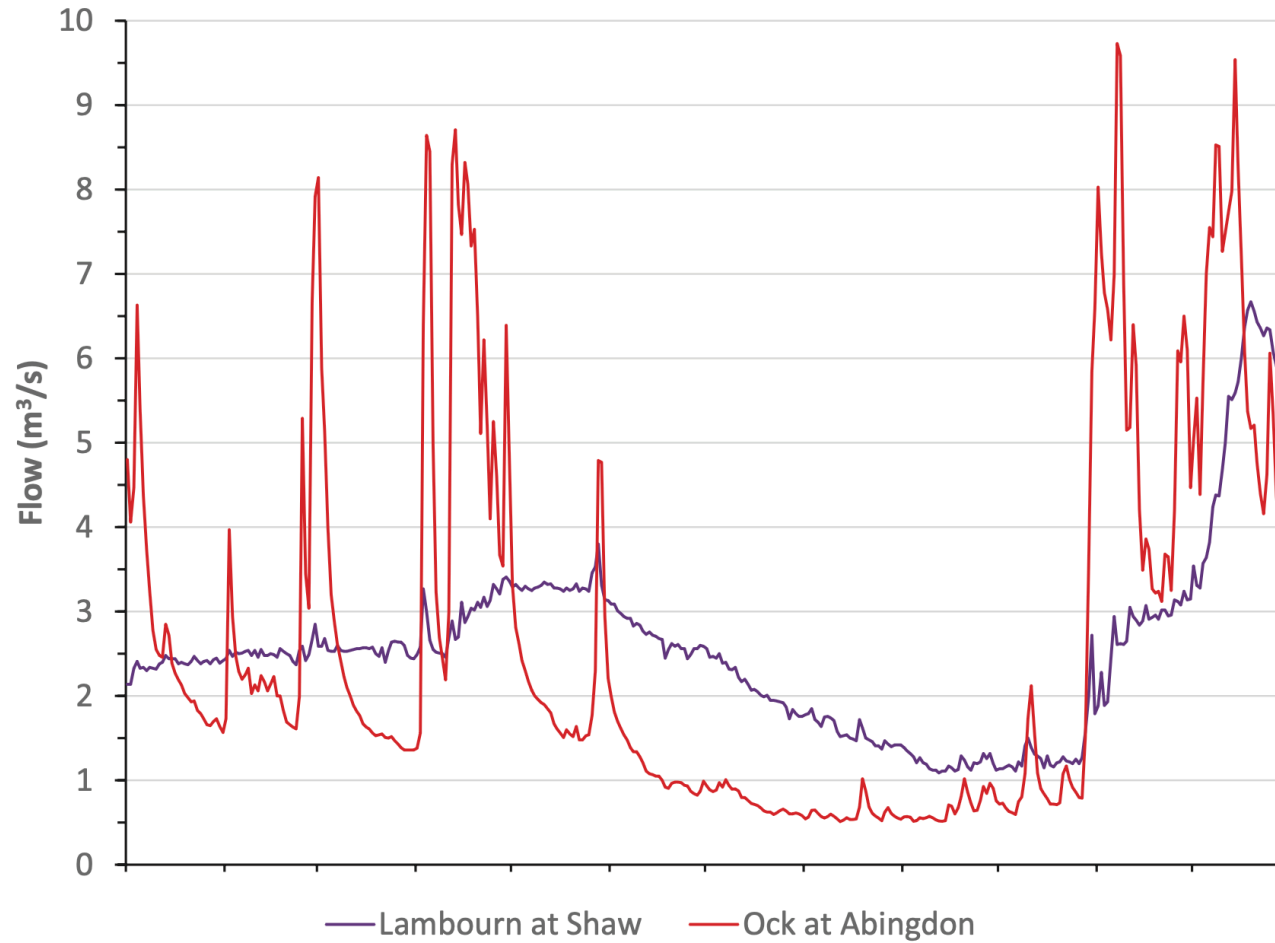


Figure 2: Comparative hydrographs for two adjacent sub-catchments in the Thames catchment.

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Catchments and their geology

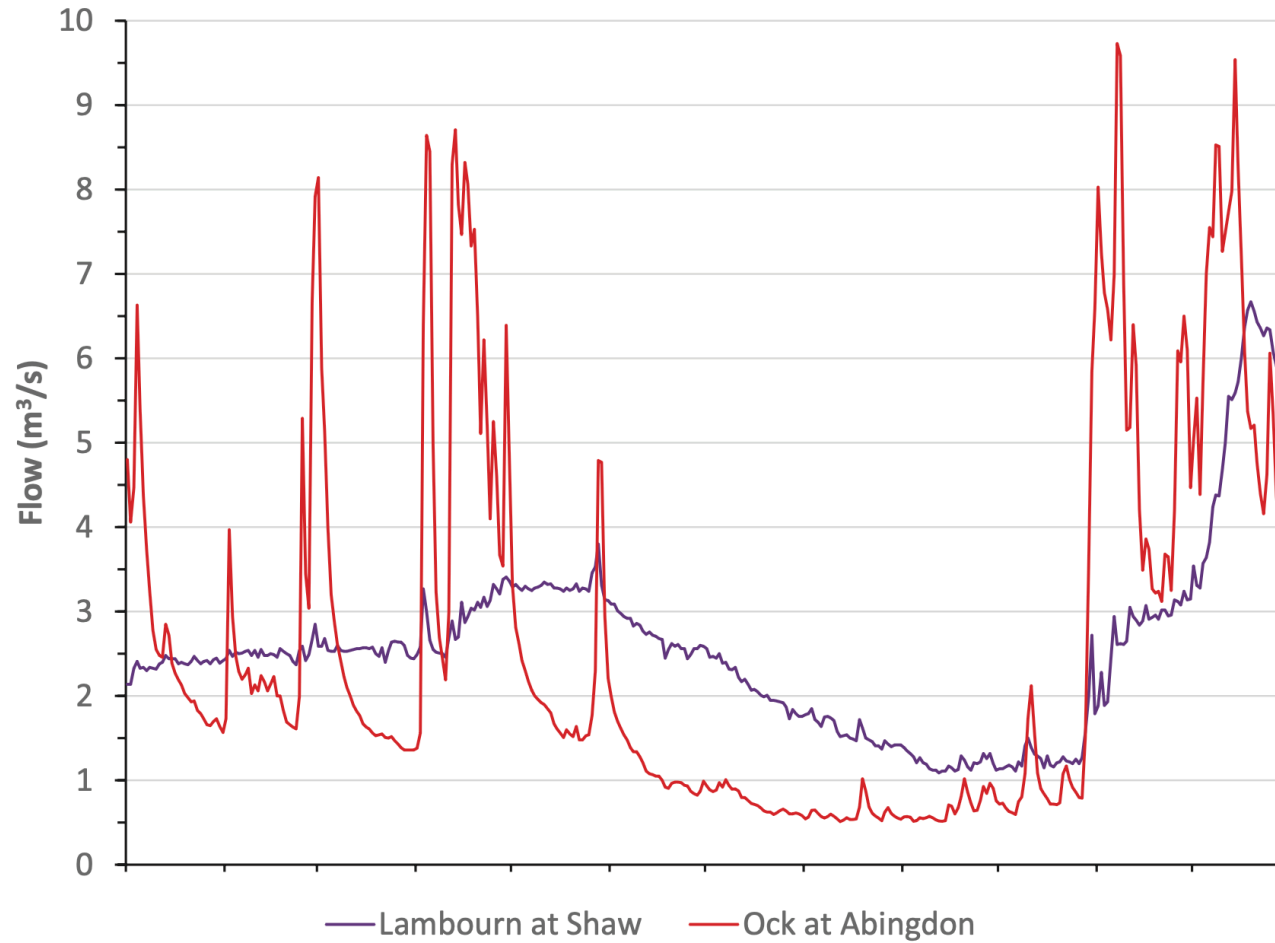
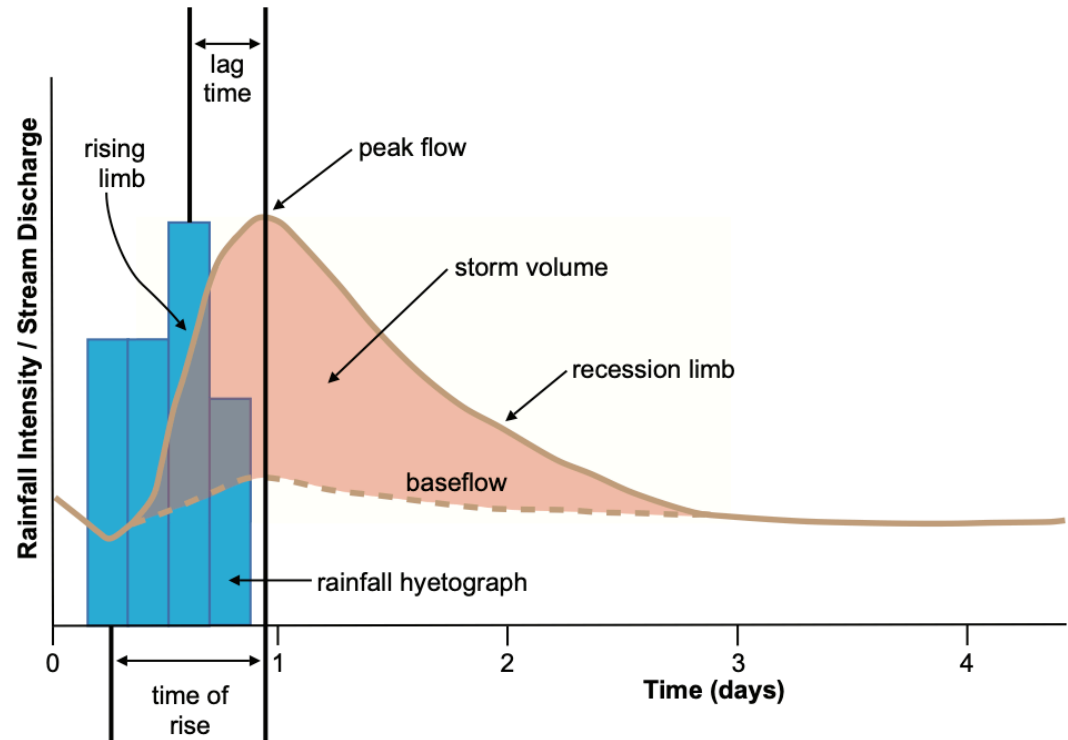


Figure 2: Comparative hydrographs for two adjacent sub-catchments in the Thames catchment.

Identical climatic conditions, but different geology. Rocks of the Lambourn are all high permeability, while the Thames catchment is mostly low permeability + bedrock.

Storm hydrograph in detail

- Rising limb: Steep section leading up to peak flow.
- Contributed by: **Channel precipitation** (rain falling directly into the stream) and rapid runoff.
- Recession limb: Follows the peak and shows a gradual decline in streamflow.
- Influenced by stormwater arriving from upstream and contributions from subsurface flow.



The exact proportion of precipitation that becomes channelized flow is difficult to determine.

Figure 3: A typical storm hydrograph

Runoff Mechanism

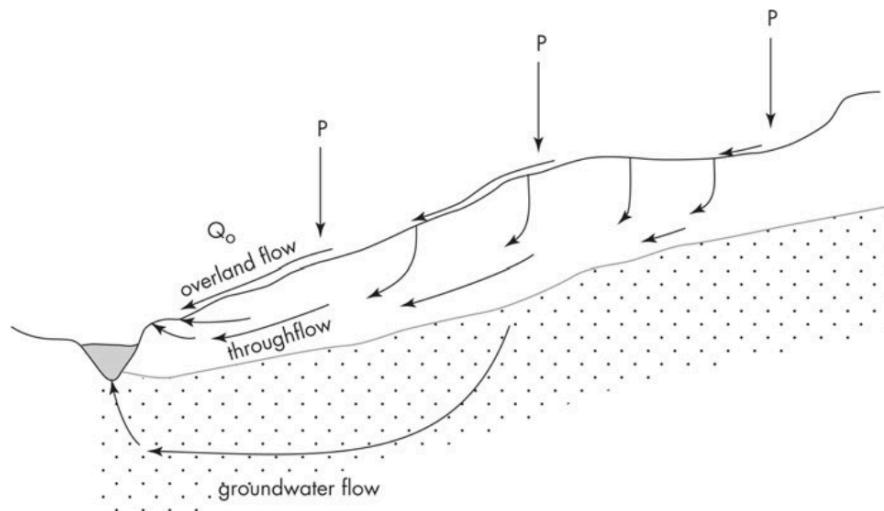


Figure 4: Hillslope runoff processes

- **Overland flow:** Water flowing over the land surface after precipitation.
- Horton (1933): Overland flow occurs when rainfall intensity exceeds the infiltration capacity of the soil. Excess water then flows as a shallow surface layer. Infiltration capacity acts as a **threshold**.
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Runoff Mechanism

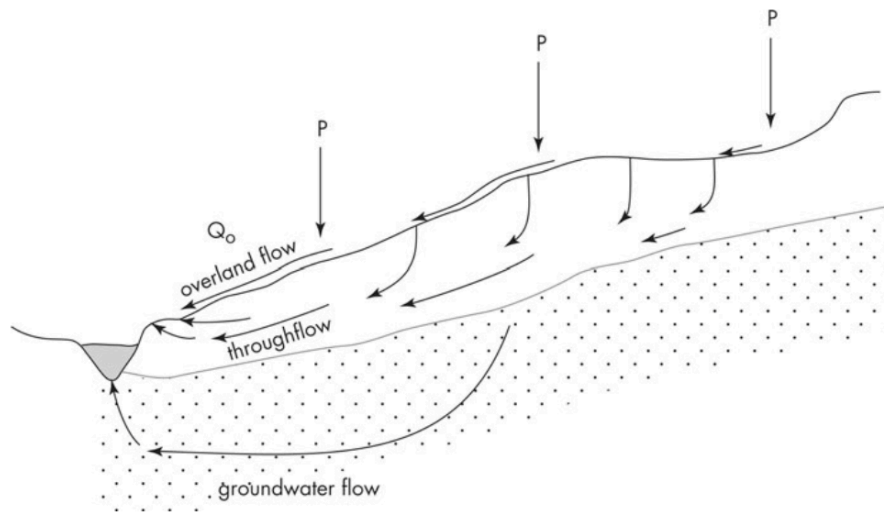


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Continuous sheets of overland water flow are rarely observed in nature.

Runoff Mechanism

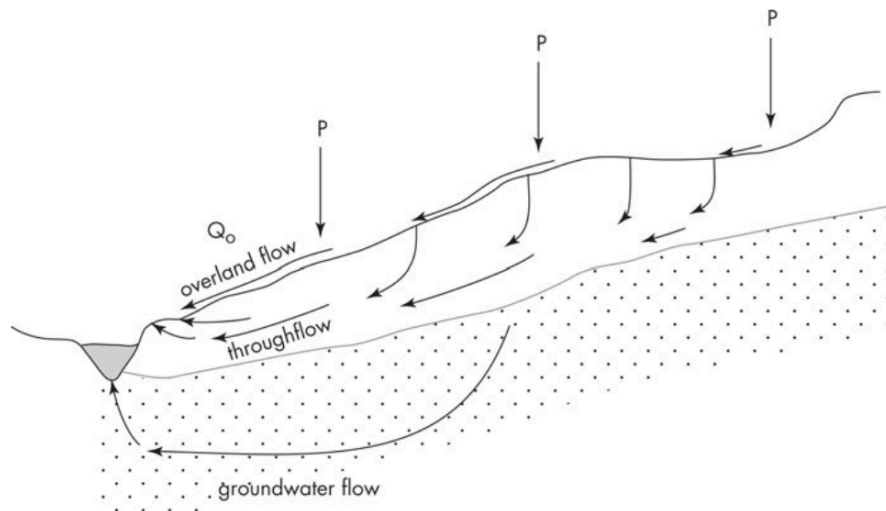


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- However, in most cases, infiltration rates exceed typical rainfall intensities.

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Runoff Mechanism

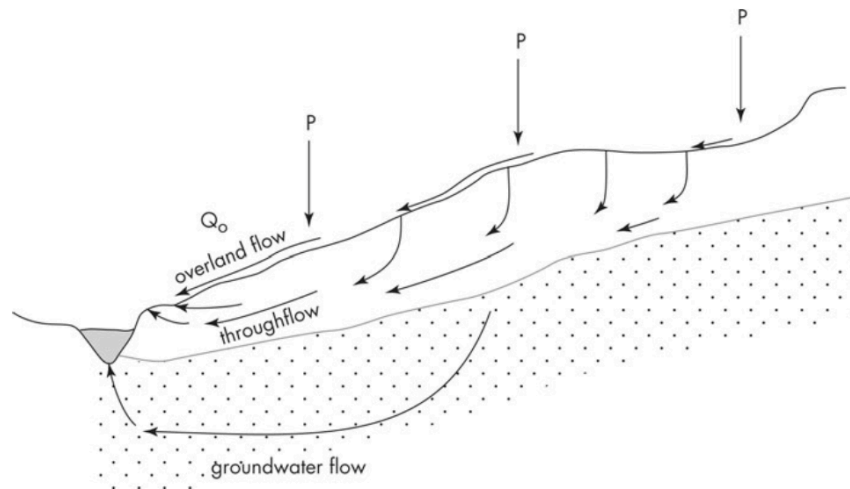


Figure 5: Hillslope runoff processes

Continuous sheets of overland water flow are rarely observed in nature.

- However, in most cases, infiltration rates exceed typical rainfall intensities.}

Soil and vegetation	Infiltration rate (mm/hr)	Rainfall type	Rainfall intensity (mm/hr)
Forested loam	100–200	Thunderstorm	50–100
Loam pasture	10–70	Heavy rain	5–20
Sand	3–15	Moderate rain	0.5–5
Bare clay	0–4	Light rain	0.5

Table 1: Typical infiltration rates versus rainfall intensities Burt (1

Runoff Mechanism

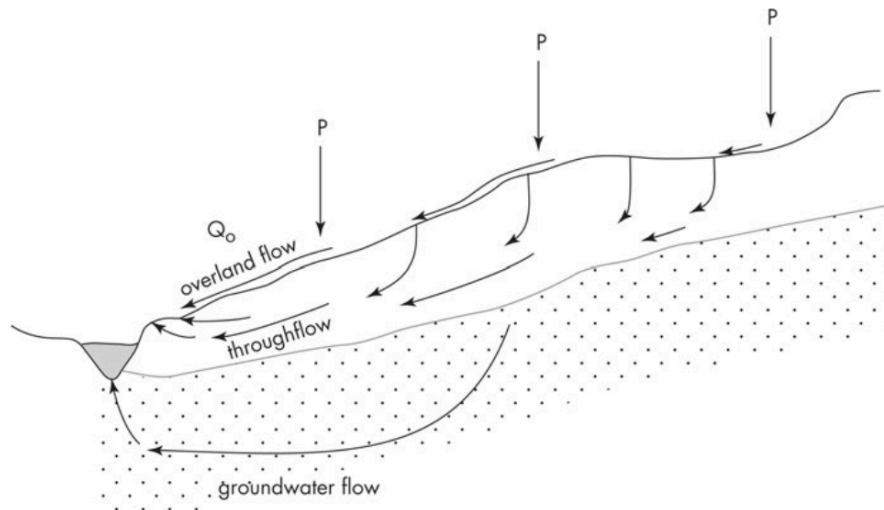


Figure 6: Fig. 5.3 Hillslope runoff processes

- Betson (1964): Introduced the concept of partial contributing areas — only specific parts of a catchment produce overland flow. The importance of infiltration is retained, but the idea of uniform thin flow is questioned.
- Hewlett & Hibbert (1967): Overland flow occurs when the water table rises to the surface due to infiltration and throughflow. This results in saturated areas near channels and lower slopes where **return flow** and **direct precipitation** dominate.

Who was right?

Ideas on stormflow

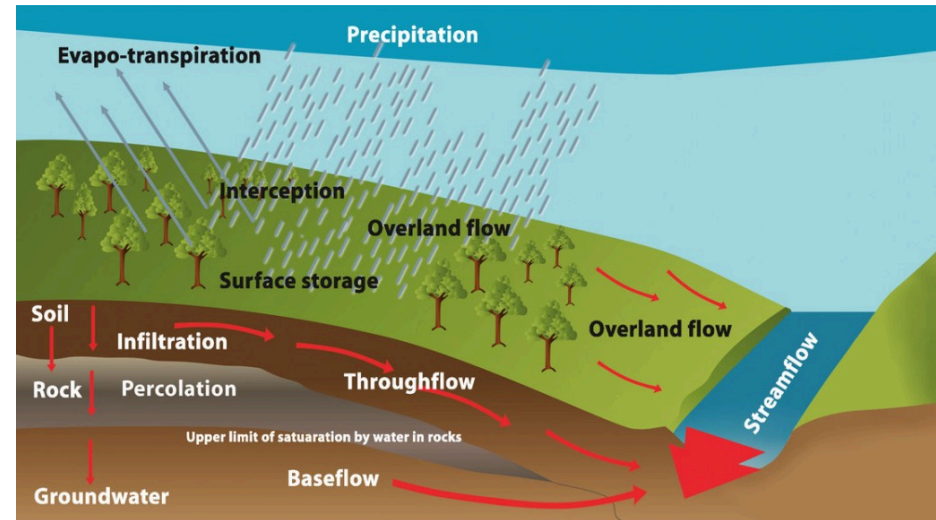
- All theories capture aspects of the truth.
- Saturated overland flow dominates in humid, mid-latitude regions.
- The **variable source area** concept best represents stormflow processes.
- In arid/semi-arid zones, intense rainfall and low infiltration lead to **flash flooding**.
- Hydrophobic soils may exhibit **nonlinear infiltration**: initially very low, then increasing.

	Horton	Betson	Hewlett and Hibbert
Infiltration	Governs overland flow	Governs overland flow	All rainfall infiltrates initially
Overland flow mechanism	Infiltration-excess	Infiltration-excess	Saturated overland flow
Contributing area	Uniform across the catchment	Confined to specific zones	Varies in time and space

Table 2

Subsurface flow

- The variable source area model proposes that only parts of the catchment contribute to the hydrograph.
- Overland flow alone can't explain total discharge observed in hydrographs.
- The gradual decline in the hydrograph's recession limb suggests a **subsurface flow** component.
- Tracer studies show significant presence of *older water* during stormflow.
- This points to **throughflow** and **groundwater contributions**.

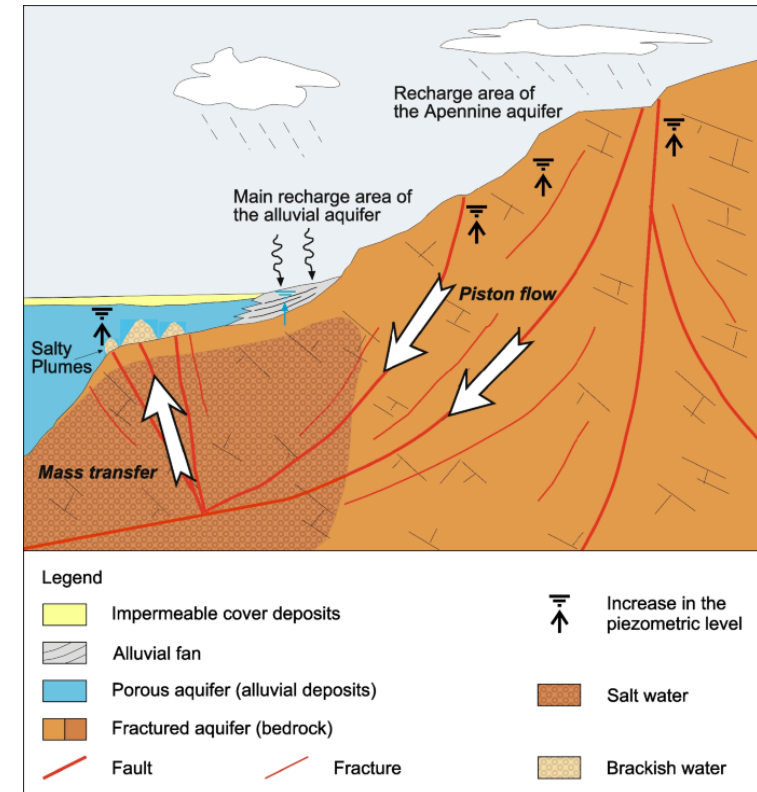


Throughflow

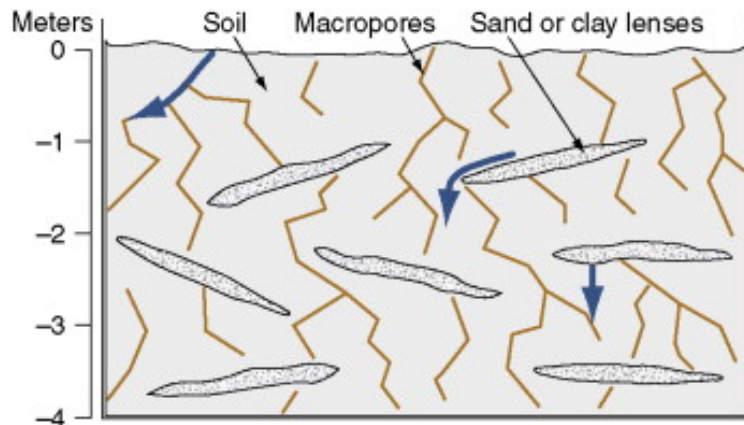
- **Throughflow:** Movement of water through the unsaturated zone (soil matrix).
- Described by the Darcy or Richards equations.
- Flow is not only vertical; on slopes, water moves laterally downslope.

Piston Flow

- Proposed by Horton & Hawkins (195): Water entering at the top of the soil column displaces older water at the bottom, which then enters the stream.
- Similar to a piston where top pressure pushes fluid at the bottom.
- The discharge can be viewed as a pressure wave.
- Though unconventional, the combination of rainfall above and impermeable bedrock below creates this **piston effect**.
- The added water generates a hydraulic gradient that drives water movement along the base of the soil layer, which typically has higher conductivity.
- Analogy: A thatched roof—water moves quickly along aligned channels (macropores), especially in the downslope direction.



Macropores



Macropores

Large, interconnected pores in soil, generally >3 mm in diameter.

- **Key pathway** for rapid subsurface flow.
- **Hydrological role:** Enable fast water movement bypassing much of the soil matrix.
- Ongoing debate: Whether these pores form continuous flow networks or are limited by disconnected segments.

Groundwater contribution

- Historically, groundwater was thought to only contribute to baseflow.
- Farvolden (1979) introduced the **capillary fringe hypothesis**.
- Concept: A “groundwater ridge” forms near the stream during recharge.
- Added infiltration saturates the unsaturated zone, generating hydraulic gradients that drive discharge to the stream.
- The water-saturation relationship in soils is **nonlinear**.
- This mechanism increases both the rate and volume of discharge.
- The Maimai experiments confirmed groundwater as a **rapid** source of streamflow.

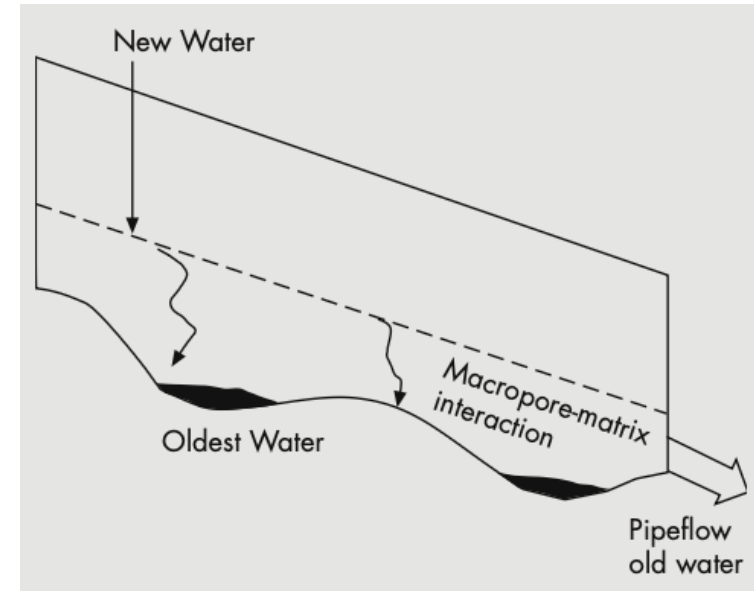
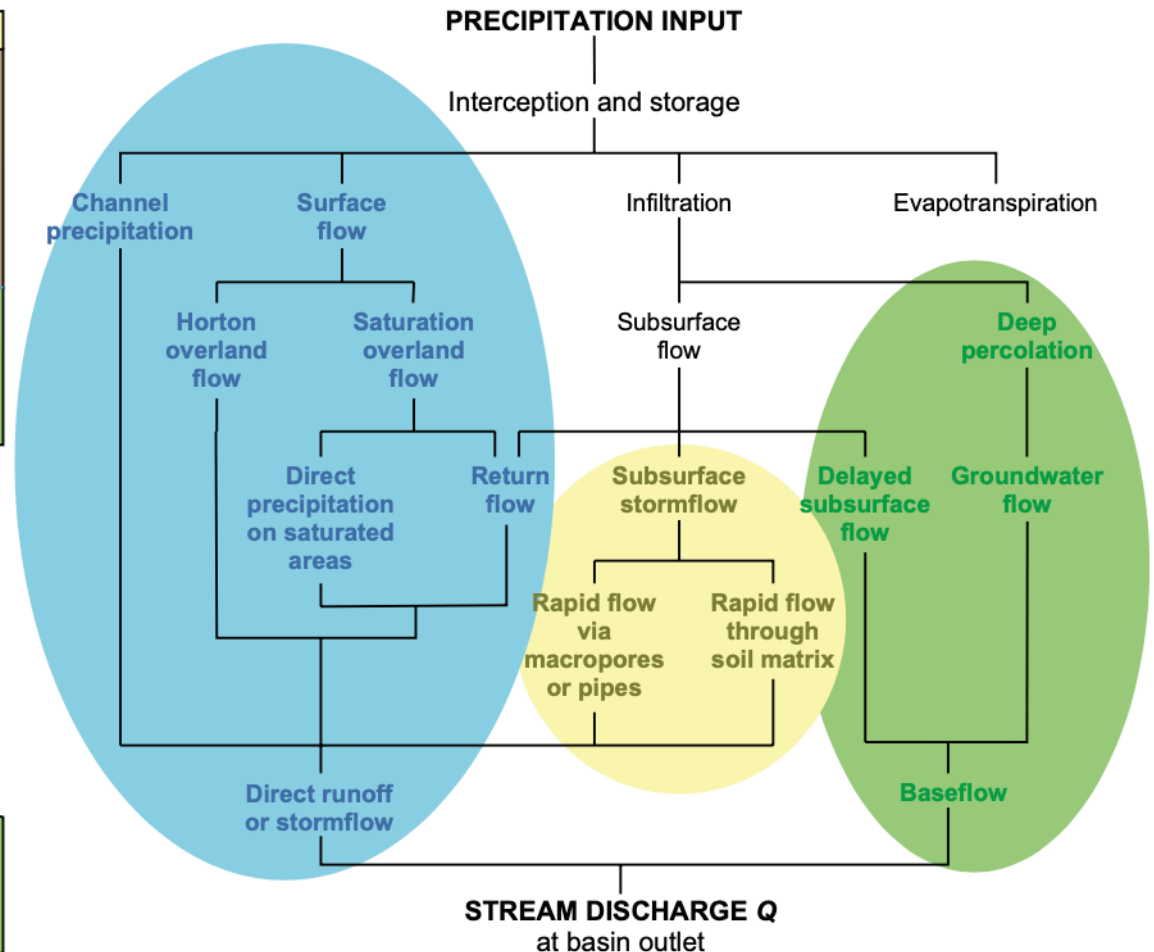
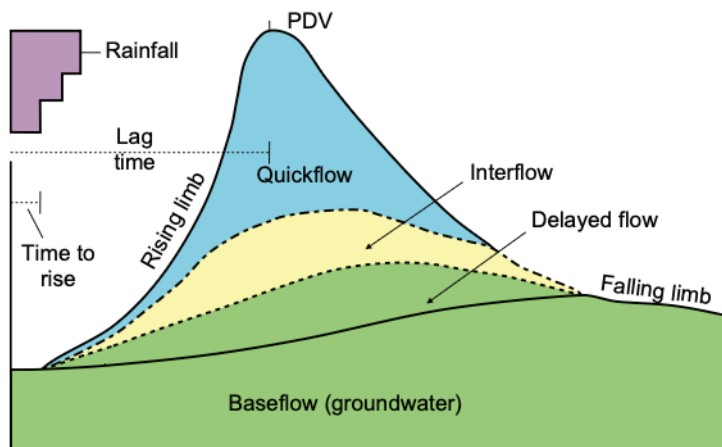
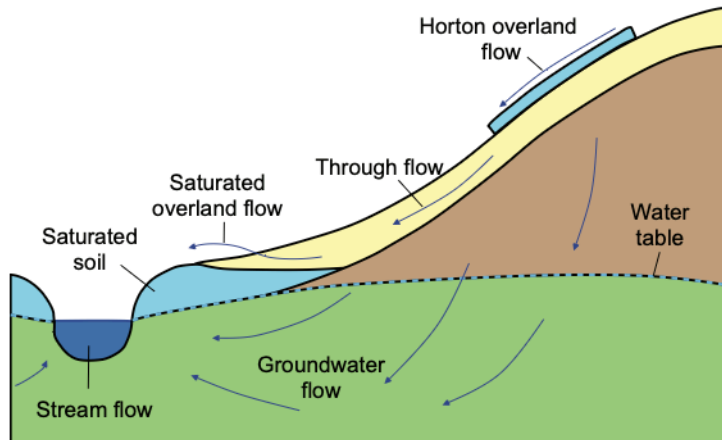


Figure 7: McGlynn et al. (2002) confirmed significant groundwater contributions to streamflow at Maimai, including piston flow and macropore evidence

A summary of different processes



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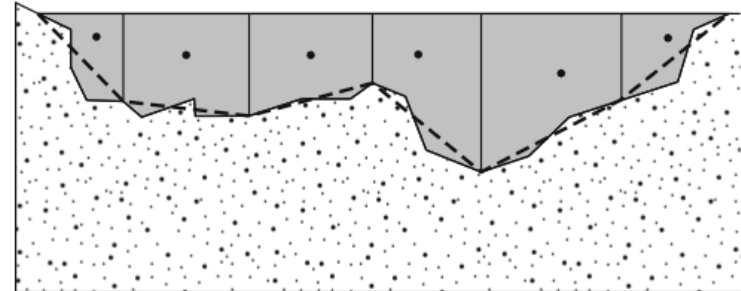
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Measuring streamflow

Velocity-area method

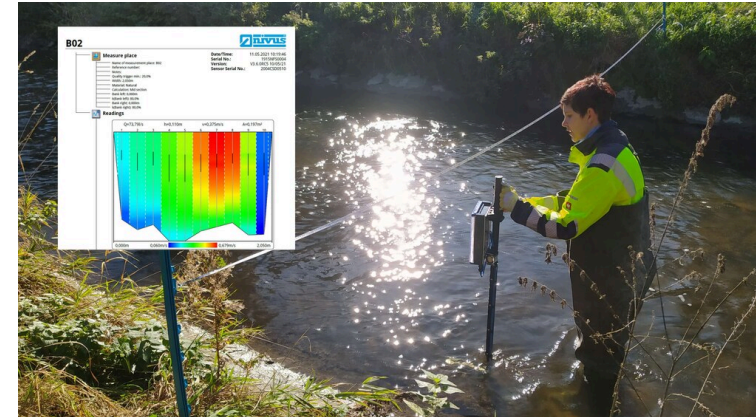
- The velocity–area method multiplies stream velocity by cross-sectional area to estimate discharge.
- Black dots: locations of velocity readings.
- Dashed lines: triangular/trapezoidal cross sections.



The velocity–area method of streamflow measurement. The black circles indicate the position of current meter velocity readings. Dashed lines represent the triangular or trapezoidal cross-sectional area through which the velocity is measured.

Velocity–area method: considerations

- The **velocity–area method** requires the assumption that the velocity measured is representative of the entire cross-sectional flow.
- Since multiple measurements across the depth are rarely feasible, adjustments are needed to account for velocity variations.
- Water flows faster near the surface than near the bed due to **bed friction**.
- A general guideline: measure velocity at **60% of depth** from the surface (or 40% above the bed).
- For deeper rivers, average readings at **20% and 80% of depth** for better accuracy.
- If no velocity meter is available, a **float method** can provide rough estimates by timing surface travel over a measured distance.
- Surface floats **overestimate** true velocity since they ride faster-flowing surface water.



- The velocity–area method works well for **small rivers**, but its **accuracy depends on sampling strategy**.
- It becomes less reliable in **turbid, shallow, or rough-bedded streams** (e.g., mountain creeks).
- In such cases, consider alternative methods like **dilution gauging**. *Illustration of float-based surface velocity measurement*

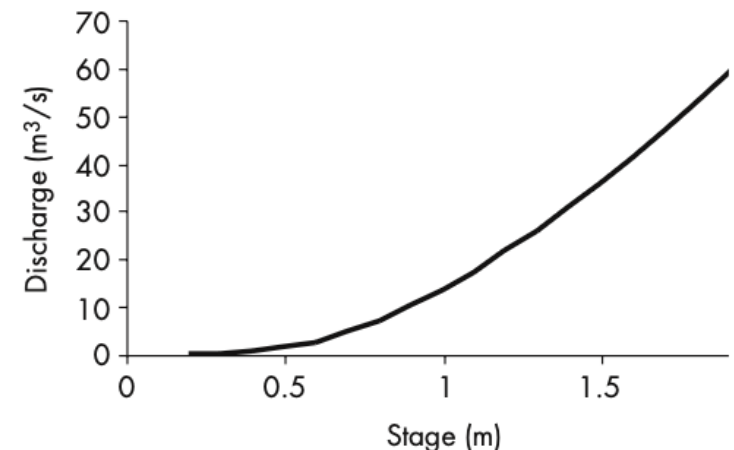
Stage–Discharge Relationship

- **Stage** = water level or height at a specific point in a river.
- **Discharge** = volumetric flow rate (e.g. $\frac{m^3}{s}$).
- Repeated discharge measurements (via velocity–area method) enable creation of a **rating curve**.
- A rating curve relates stage to discharge and allows **continuous discharge estimation** from simple stage measurements.
- Developed by pairing stage readings (e.g. using a **stilling well**) with discharge data.
- Rating curves are typically **non-linear**, reflecting the geometry of riverbanks:
As rivers fill between banks, **more water is required to raise stage** than at low flows.
- Assumes a **stable riverbed**. Changes (e.g. due to flood scouring or deposition) will invalidate the curve.
- This is why **flumes or weirs** are often installed to stabilize the control section.

Limitations:

- Few high-flow measurements → less reliable for peak floods.
- High-flow events are dangerous and infrequent.
- Thus, **errors increase at the upper end** of the curve.

Remember: discharge is **inferred** from measured stage — not directly observed.



Flumes and weirs

- In principle the same as stage-discharge relationship, but with a control section.
- Flumes and weirs are stream **gauging structures** that extend the **stage–discharge** method by imposing control over stream velocity and cross-sectional area.
- By doing so, they provide a **continuous record** of stream discharge.
- The key is to **standardize flow** through a known cross-sectional area so that velocity becomes predictable or known.
- This is especially useful because stream velocity varies with depth, but gauging structures enforce a **consistent flow regime**.
- These structures are designed so that discharge becomes a function of **stage** alone.



Discharge equations for weirs

- **Discharge** through weirs is determined by known formulas based on weir shape.
- For a **V-notch weir** (Very accurate for low-flow rates):

$$Q = 0.53\sqrt{2g} C \tan\left(\frac{\theta}{2}\right) b^{2.5}$$
- Where:
 - Q = discharge (m^3/s)
 - g = gravity (9.81 m/s^2)
 - C = discharge coefficient (depends on θ)
 - θ = notch angle
 - b = stage height (m)
- For a 90° V-notch, $C = 0.578$, and: $Q = 1.366 b^{2.5}$
- These equations follow **ISO 1438** standards for gauging structures.

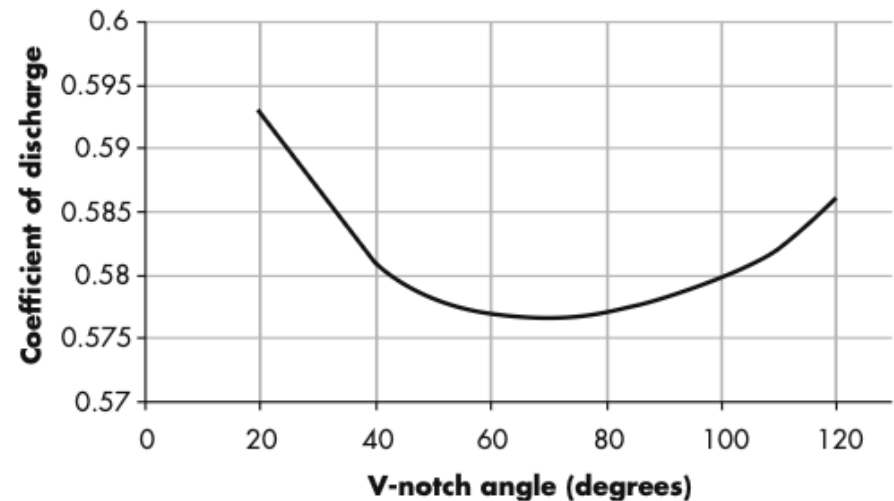
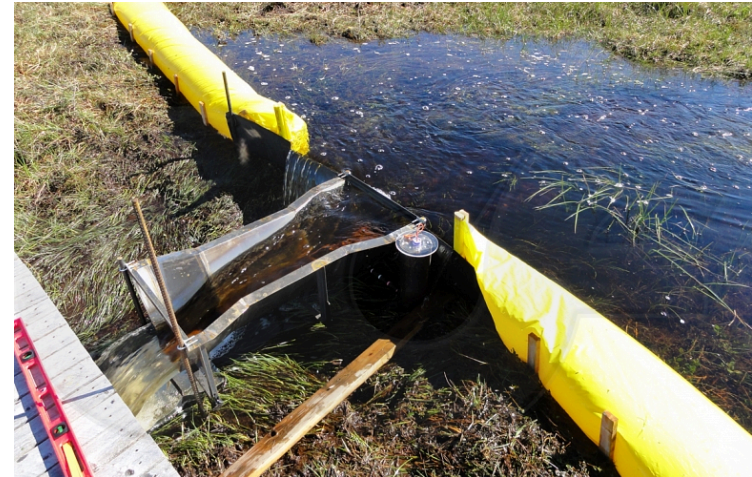


Figure 8: Discharge coefficient for different V-notch angles

Flumes and weirs: design considerations

- The **shape** of the gauging structure affects sensitivity to changes in flow:
 - V-notches are excellent for **low flows** due to higher sensitivity.
 - As discharge increases, cross-sectional area expands nonlinearly.
- Under **high flows**, it's crucial that water doesn't bypass the structure.
- V-notches are commonly used with **90° or 120° angles** based on application needs.
- **Stilling ponds** slow down water before measurement but can trap sediment.



A trapezoidal flume helps flush sediment

Flumes vs. weirs

- **Flumes and weirs** both allow continuous measurement of stream discharge but have **key differences**:
 - **Weir**: Water flows **over** a structure (like a small waterfall).
 - **Flume**: Water flows **through** a structure without dropping.
- Flumes tend to produce **less sedimentation**, making them suitable for **high-energy environments**.
- Trapezoidal flumes can **flush out sediment** efficiently, avoiding stilling pond build-up.
- Both are suitable for small streams but not large rivers, due to structural stress.



Summary

So here we covered that:

- Runoff is the movement of water toward channels, influenced by rainfall, soil, vegetation, and terrain.
- Stormflow is shaped by infiltration capacity, contributing areas, and subsurface dynamics.
- Horton, Betson, and Hewlett & Hibbert offer complementary theories explaining overland and saturated flow.
- Key mechanisms include variable source areas, piston flow, and macropore-driven throughflow.
- Groundwater contributes rapidly via capillary rise during storms.
- Streamflow is measured using methods like velocity–area, stage–discharge curves, and gauging structures (weirs/flumes).